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Conception and Prototypical Implementation of a System for Synthesizing Singing Voices from Text and Melody

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Abstract

Automated speech synthesis is now widely used. Producing a singing voice adds further requirements, because musical information such as pitch, note duration and rhythm has to be taken into account alongside the text. This study thesis examines and compares approaches to singing-voice synthesis, in particular how different ways of processing the linguistic and musical information affect the resulting audio signal.

One of these approaches is then implemented as a prototype and applied in the form of a singing robot.

Declaration of Originality

Pursuant to Section 20 paragraph 5 of the Study and Examination Regulations for the bachelor's and master's programmes of the Baden-Wuerttemberg Cooperative State University (DHBW StuPrO) of 7 March 2024.

I hereby declare that I have written this thesis independently and have used no sources or aids other than those indicated, that I have not submitted this thesis for any other examination with the same or comparable content, and that it has not been published before.

Offenburg, <Datum> signed, Jannis Saur

Place, date Signature

General remarks

For readability, this thesis uses the generic masculine. All personal designations apply equally to every gender. Some technical terms from software development are kept in their original English form, because they are established in the field and describe the subject more precisely.

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Contents

1 Introduction **1**

2 Fundamentals **2**
2.1 Production and Description of Spoken Language 2
2.2 Distinction between Speech and Singing 3
2.3 Musical Description of the Melody 4
2.4 Digital Representation of Sound 5
2.5 From Text and Melody to the Audio Signal 6

3 Planned Approach **8**

4 Implementation **9**

5 Evaluation and Comparison **10**

6 Conclusion and Outlook **11**

7 Appendix **VI**

List of Figures

List of Listings

List of Abbreviations

API	Application Programming Interface

1 Introduction

Introduction to the topic.

2 Fundamentals

This chapter defines the terms on which the later design of the synthesis system is based. It first describes how spoken language is produced and divided into linguistic units. It then distinguishes speech from singing: a singing voice uses the same articulatory organs, but pitch and timing are set by the melody. After that, the musical quantities required for this are introduced and linked to the text. The chapter closes with the digital representation of sound and the processing chain from text and melody to the audio signal. Concrete methods and their comparison are not the subject of this chapter.

2.1 Production and Description of Spoken Language

In order to synthesise a singing voice later on, spoken language is examined first.

2.1.1 Phonation and Articulation

During speech, air is pressed out of the lungs. This acts as the energy source of spoken language. A tone is produced by the vibration of the vocal folds and by changes in the shape of the mouth.

In addition to the tone itself, speakers usually move parts of the body to support what is said. This is generally known as articulation. The body movement acts as a linguistic signal, comparable to the audio signal of spoken language. Together with the spoken words, this forms an overall picture that lets the listener reconstruct the message **anderson2022essentials**.

- Lungs and the outward airflow as the energy source; subglottal pressure.
- Phonation: vibration of the vocal folds, and from that the fundamental frequency.
- Articulation: tongue, lips, jaw and velum change the shape of the vocal tract and therefore the sound.
- Voiced and unvoiced sounds: periodic excitation versus noise excitation.
- Kept short: the listener's hearing is not modelled. What matters is how the signal is produced, not how it is perceived in the inner ear.

Articulation, phonation and the difference between voiced and unvoiced sounds follow **anderson2022essentials**. The fundamental frequency as a result of vocal-fold vibration is described by **jurafsky2026slp**.

2.1.2 Source-Filter Model

- Source: periodic or noise-like excitation at the glottis.
- Filter: resonances of the vocal tract, described by formants.
- Separating excitation from timbre is a shared mental model of many synthesis methods.
- Limit of the model: it explains vowels well, but consonants and transitions only in a simplified way.

The source-filter model and the formants follow **fant1981sourcefilter** and **jurafsky2026slp**. The technical separation of excitation and vocal-tract filter in speech processing is described by **backstrom2022speech**.

2.1.3 Linguistic Units

- Sound, phoneme and syllable; the syllable is the unit to which notes are later attached.
- Relation between spelling and pronunciation (grapheme to phoneme), because the input is text while the signal needs sounds.
- Prosody: linguistic control of stress, intonation and speaking rate.
- In singing, this prosody is not estimated freely. It is replaced or overlaid by the melody. That motivates the next section.

Phoneme, syllable and stress follow **anderson2022essentials**. The relation between spelling and the sound sequence, and pitch as an expression of intonation, is described by **jurafsky2026slp**.

2.2 Distinction between Speech and Singing

Speech and singing are produced with the same apparatus. This section records which properties are shared and which ones the synthesis has to control in addition.

2.2.1 Shared Basis

- The same articulatory apparatus and the same source-filter principle.
- Intelligibility depends on how well sounds can be distinguished, above all vowels, consonants and the transitions between them.

- Singing is not a second language. It is a different use of the same sounds in time and pitch.

The shared articulatory apparatus, the formants and the ways in which singing differs from the speaking voice follow **sundberg1979singing**.

2.2.2 Pitch and Temporal Organisation

- Speech: pitch follows intonation. It is relative and changes continuously.
- Singing: pitch follows discrete scale steps and is held for the duration of the note.
- Speech: syllable duration comes from prosody and speaking rate.
- Singing: duration and onset follow metre, tempo and note values.
- Vowels are lengthened, consonants stay short. That determines how text can be distributed across notes.

Pitch, formant tuning and phrasing in the comparison of speech and singing follow **sundberg1979singing**.

2.2.3 Timbral Properties of the Singing Voice

- Vibrato, legato and gliding pitch transitions as features that distinguish a singing voice from isolated tones.
- Loudness contour within a note.
- Which of these features are needed for intelligibility, and which only increase naturalness. The priority is set in the design; here they are only named.

Vibrato, phonation and loudness of the singing voice follow **sundberg1979singing**.

2.3 Musical Description of the Melody

The melody is the second input of the system. Only quantities that occur as control data are described: pitch, duration and their assignment to the text. Harmony and organology are left aside.

2.3.1 Pitch, Duration and Rhythm

- Pitch as a frequency and as a musical scale step, including the octave.

- Note duration, rest, metre and tempo.
- Relation between the fundamental frequency of the voice and the notated pitch.
- Only as much about tuning (equal temperament) as is needed to turn a note into a frequency.

Notated pitch, duration, rests, metre and tempo follow **gotham2023openmusic**. The relation between notated pitch and fundamental frequency is described by **jurafsky2026slp**.

2.3.2 Alignment of Text and Melody

- Usual case: one syllable per note.
- Melisma: one syllable across several notes.
- Several syllables tied in time to one longer note.
- Why text and melody are separate inputs and have to be aligned before synthesis.

The syllable as a textual unit is described by **anderson2022essentials**. Text and melody as the joint input of a singing synthesis follow **cho2021svs**.

2.4 Digital Representation of Sound

The result of the synthesis is a digital audio signal. This section defines the representations in which a singing voice is described and generated.

2.4.1 Signal and Spectrum

- Sound as the change of sound pressure over time; sampling and sampling rate.
- Frequency range that matters for a singing voice: fundamental frequency and overtones.
- Short-time spectrum and spectrogram as a time-resolved representation.
- Why many methods do not generate the waveform directly, but a spectral representation from which the signal is reconstructed.

Sampling, the short-time spectrum and the spectrogram follow **backstrom2022speech**. Waveform, spectrum and the speech spectrogram are described in addition by **jurafsky2026slp**.

2.4.2 Acoustic Features of the Singing Voice

- Contour of the fundamental frequency.
- Spectral envelope as the carrier of timbre and of phone identity.
- Voiced versus unvoiced, including short consonantal segments.
- These features as the target quantities between the symbolic input (text, notes) and the audible signal.

Fundamental frequency, formants, voiced and unvoiced excitation and the spectral envelope follow **jurafsky2026slp**.

2.5 From Text and Melody to the Audio Signal

The previous sections come together here. The processing chain is defined without choosing a concrete method yet.

2.5.1 Speech Synthesis and Singing Synthesis

- Speech synthesis: speech is produced from text, and pitch and duration are estimated from the prosody.
- Singing synthesis: text and melody are given. Pitch and duration are inputs, not estimated values.
- Consequence for this thesis: a speech-synthesis system does not cover the task, because the musical control is missing.

Speech synthesis from text, in which the prosodic contours are generated, is described by **backstrom2022speech**. That a singing synthesis takes pitch and rhythm from the musical score follows **cho2021svs**.

2.5.2 Processing Pipeline

- Input: text in written form or as a sound sequence, and the melody as a sequence of note events.
- Linguistic analysis and assignment of syllables to notes.
- Control quantities: sound sequence, fundamental frequency, duration, optionally loudness.

- Acoustic model: the control quantities become a spectral or direct representation of the signal.
- Waveform synthesis: this produces the audio signal that can be played back.
- Which methods implement these steps, and which of them is built as a prototype, follows in the planned approach and in the evaluation.

The chain of text analysis, acoustic features and waveform synthesis follows **backstrom2022speech**. The corresponding chain for singing, from the score through an acoustic model to the waveform, is described by **cho2021svs**.

3 Planned Approach

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4 Implementation

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5 Evaluation and Comparison

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6 Conclusion and Outlook

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7 Appendix

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